

Box-Office Revenue Prediction Using Elastic Net Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

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by

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DEPARTMENT OF

CSE

Name of Title: **Box Office Revenue Prediction Using Elastic Net Regression for Sports and Entertainment Industry**

1. Abstract:

The entertainment industry has experienced tremendous growth over the past decade, with movies generating billions of dollars in revenue worldwide. Predicting a movie's box-office revenue before its release has become an important task for producers, distributors, investors, and marketing teams. Accurate predictions help in budget planning, promotional strategies, theatre allocation, and financial risk management. However, box-office performance depends on multiple factors such as production budget, genre, cast popularity, director reputation, release season, runtime, audience ratings, critic reviews, and marketing campaigns. These variables often have complex relationships and multidisciplinary, making prediction a challenging machine learning problem.

This project presents a **Box Office Revenue Prediction System using Elastic Net Regression**, a supervised machine learning algorithm that combines the strengths of both Lasso (L1) and Ridge (L2) regularization techniques. Elastic Net effectively handles correlated features, reduces over fitting, performs automatic feature selection, and improves the prediction accuracy compared to traditional linear regression models.

The project begins with collecting historical movie data from publicly available datasets such as Imbed, TMDb, or Kaggle. The dataset undergoes preprocessing steps including data cleaning, handling missing values, encoding categorical variables, feature scaling, and feature engineering. The processed data is then divided into training and testing sets. The Elastic Net Regression model is trained on the datasets and evaluated using performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 Score.

The developed model enables stakeholders to estimate expected box-office revenue before movie release, thereby assisting in strategic decision-making. The proposed system demonstrates that Elastic Net Regression is an efficient and reliable technique for revenue prediction in the sports and entertainment domain, providing improved accuracy, better interpret ability, and robust performance on real-world datasets.

2. INTRODUCTION: The movie industry is one of the largest sectors in the global entertainment market. Every year, thousands of films are released, involving huge investments in production, marketing, and distribution. Despite careful planning, predicting the financial success of a movie remains a difficult challenge due to the influence of numerous interconnected factors.

Machine Learning has emerged as a powerful solution for predictive analytics by learning patterns from historical data. Among various regression algorithms, **Elastic Net Regression** is particularly suitable because it combines the advantages of Ridge and Lasso Regression. It not only minimizes prediction errors but also selects the most important features while handling multidisciplinary effectively.

This project aims to develop a predictive model capable of estimating box-office revenue using movie-related attributes such as production budget, genre, release date, cast popularity, director experience, runtime, critic ratings, and audience reviews.

The project follows a complete machine learning pipeline consisting of data collection, preprocessing, exploratory data analysis, feature engineering, model training, evaluation, and prediction. The final model helps movie producers, investors, distributors, and production houses make informed business decisions by estimating potential revenue before a movie's release.

3. SOFTWARE REQUIREMENTS:

- **Operating System:** Windows 10/11
- **Programming Language:** Python 3
- **IDE:** Jupiter Notebook or VS Code
- **Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib
- **Dataset:** Kaggle Movie Dataset
- **Machine Learning Model:** Elastic Net Regression



Signature of the Guide



Course Instructor